MERIDIAN TRUST BANK: PREDICT. PRIORITISE. RETAIN.

The Challenge: Build an AI-Powered Customer Retention Decision Engine

Meridian Trust Bank is facing customer attrition that could affect its deposits, customer relationships and future revenue. The bank has a simulated dataset of 10,000 retail banking customers containing demographic, financial, behavioural, digital engagement and relationship-management information, along with a historical churn outcome.

The bank does not simply want to know which customers have left. It wants to identify which existing customers may be at risk of leaving, understand the signals associated with that risk, and determine where limited retention resources should be deployed.

Your mission: Using the supplied customer dataset, build a reproducible analytics solution that combines exploratory data analysis, machine learning and business decision-making.

Your task is to prepare a recommendation for the bank's management committee. You may use Excel, SQL, Python, Power BI, or another suitable analytics tool. State your assumptions clearly and distinguish descriptive findings, predictive estimates, and recommendations.

Your solution must:

1. Identify the strongest patterns associated with customer churn using data exploration and visualisation.
2. Build and evaluate at least one machine-learning model to estimate customer churn risk.
3. Convert model predictions into a ranked list of customers or customer groups for retention prioritisation.
4. Incorporate average monthly balance at risk, not just predicted churn probability, into your prioritisation logic.
5. Recommend targeted retention interventions and design a measurable pilot to test their effectiveness.
6. Demonstrate that your solution works on previously unseen customer records or a hidden test input.

The winning team will not necessarily have the most complex model. It will demonstrate a technically sound, reproducible and explainable solution that turns customer data into a credible management decision.

6. Deliverables

- A cleaned analysis dataset or a clearly documented data-cleaning process.
- An exploratory analysis workbook, SQL script, Python notebook, or equivalent evidence of analysis.
- At least three visualisations showing churn patterns.
- A concise management report (approximately 1,000–1,500 words) or a 5–7 slide presentation.
- A recommended retention pilot with KPIs and an explanation of how results will be evaluated.

Dataset Variable Guide

Variable(s)	Description
customer_id	Customer identifier
age, gender, region	Basic customer descriptors
customer_segment	Mass Market, Mass Affluent, Affluent, or Private Banking
tenure_years	Length of the customer relationship in years
monthly_income	Recorded monthly income
avg_monthly_balance	Average monthly balance
num_products	Number of bank products held
has_credit_card, has_loan, has_investment_product	Binary indicators for selected products
digital_engagement_score	Digital engagement score
branch_visits_last_year	Branch visits during the last year
complaint_count_last_year	Complaints recorded during the last year
credit_score	Customer credit score
relationship_manager_assigned	Whether a relationship manager is assigned (1/0)
nps_score	Net Promoter Score field in the dataset
last_transaction_days_ago	Days since the last transaction
churned	Target outcome: 1 = churned; 0 = not churned

Rules for AI, internet and coding tools

Open AI should be encouraged, but students must demonstrate that they understand and can validate their solutions.

- Permitted: ChatGPT and other AI assistants, online documentation, Python libraries, open-source models, search engines and coding assistants.
- Required: Disclose the AI tools used and describe how their outputs were verified.
- Required: The submitted code must run and produce outputs from the supplied data; screenshots or AI-generated explanations alone do not count as implementation.
- Required: Explain the target variable, feature selection, train/test split, evaluation metrics and customer-ranking logic.
- Prohibited: Hardcoding expected answers, manually fabricating model metrics, using hidden labels to create predictions, or presenting correlation as causation.
- Data handling: Use only the fictional/simulated data provided for the challenge. Do not upload any real customer or confidential banking information to external AI services.

Evaluation criterion

Business problem framing and relevance	10 M
Data quality, cleaning and reproducibility	10 M
Exploratory analysis and quality of evidence	15 M
ML implementation and validation	20 M
Customer prioritisation and business logic	15 M
Retention strategy and pilot design	10 M
Robustness, explainability and responsible AI	10 M
Demonstration, communication and technical defence	10 M

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